

1. INTRODUCTION

Power BI Exploratory Data Analysis (EDA) was performed on Shopify Stock Data to understand stock price movements and trading activity over time. Simple Moving Averages, crossover analysis, price trends, trading volume, interactive filters and custom tooltips were created using Power BI and DAX.

2. OBJECTIVES

The main objectives of Week 4 are:

- To calculate 20-Day and 50-Day Simple Moving Averages using DAX.
- To compare the moving averages using a crossover signal.
- To analyze Shopify stock price trends.
- To analyze trading volume over time.
- To identify significant volume spikes.
- To create interactive date filters.
- To create a dynamic stock price selector.
- To create a custom tooltip for detailed stock information.
- To build an interactive Shopify Stock EDA dashboard.

3. SIMPLE MOVING AVERAGE

- Simple Moving Average (SMA) is used to calculate the average value over a specific number of days. In this project, 20-Day and 50-Day moving averages were created using DAX.
- **20-Day SMA**
- The 20-Day SMA calculates the average trading volume over the previous 20-day period.
- **50-Day SMA**
- The 50-Day SMA calculates the average trading volume over the previous 50-day period.

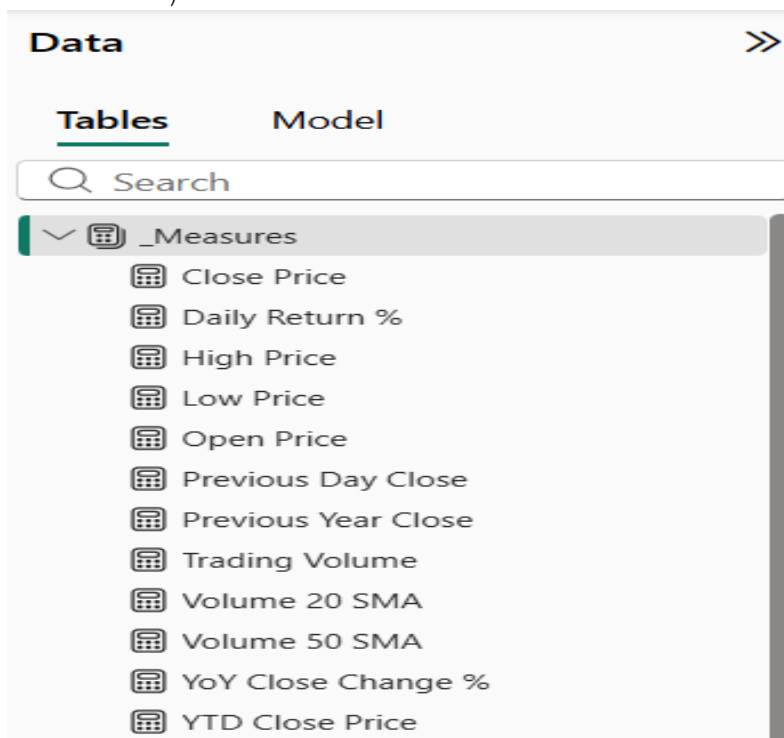
DAX Measures

```
Volume 20 SMA =  
VAR CurrentDate = MAX(Dim_Date[Date])  
VAR PeriodDays = 20  
VAR DateInPeriod =  
    DATESINPERIOD(  
        Dim_Date[Date],  
        CurrentDate,  
        -PeriodDays,
```

```

        DAY
    )
RETURN
    CALCULATE (
        AVERAGE (Shopify_Stock[Volume]),
        DateInPeriod
    )
Volume 50 SMA =
VAR CurrentDate = MAX(Dim_Date[Date])
VAR PeriodDays = 50
VAR DateInPeriod =
    DATESINPERIOD(
        Dim_Date[Date],
        CurrentDate,
        -PeriodDays,
        DAY
    )
RETURN
    CALCULATE (
        AVERAGE (Shopify_Stock[Volume]),
        DateInPeriod
    )

```



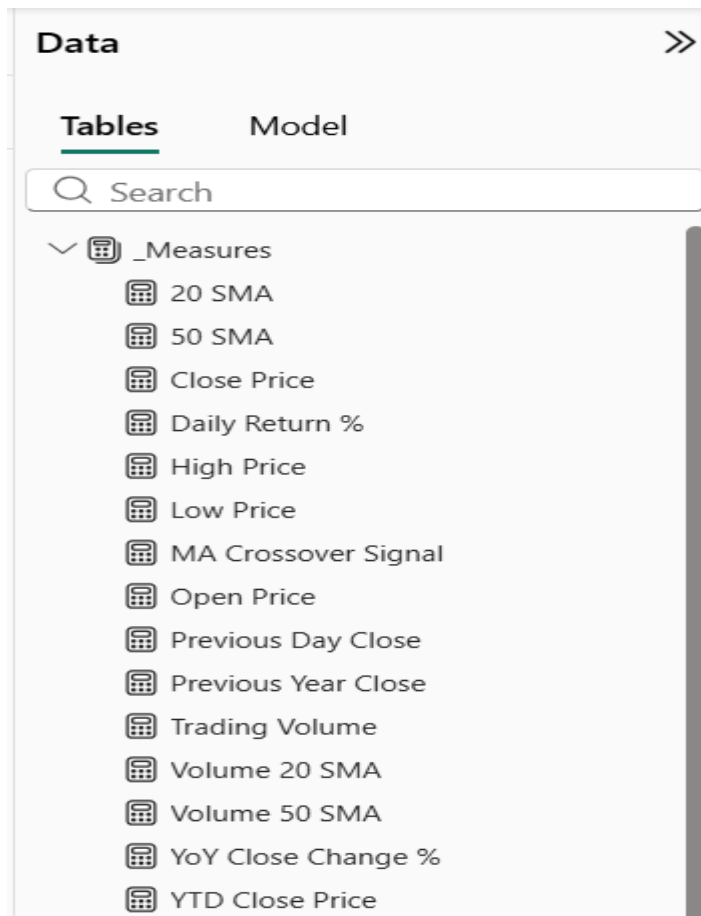
4. MOVING AVERAGE CROSSOVER SIGNAL

A Moving Average Crossover Signal was created by comparing the 20-Day SMA with the 50-Day SMA.

When the 20-Day SMA is greater than the 50-Day SMA, the signal displays Bullish ($20 > 50$). When the 20-Day SMA is lower than the 50-Day SMA, the signal displays Bearish ($20 < 50$).

DAX Measure

```
MA Crossover Signal =  
IF(  
    ISBLANK([20 SMA]) ||  
    ISBLANK([50 SMA]),  
    BLANK(),  
    IF(  
        [20 SMA] > [50 SMA],  
        "Bullish (20 > 50)",  
        "Bearish (20 < 50)"  
    )  
)
```



5. STOCK PRICE TREND ANALYSIS

A price trend chart was created to analyze Shopify stock prices over time. The chart contains Close Price, 20-Day SMA and 50-Day SMA.

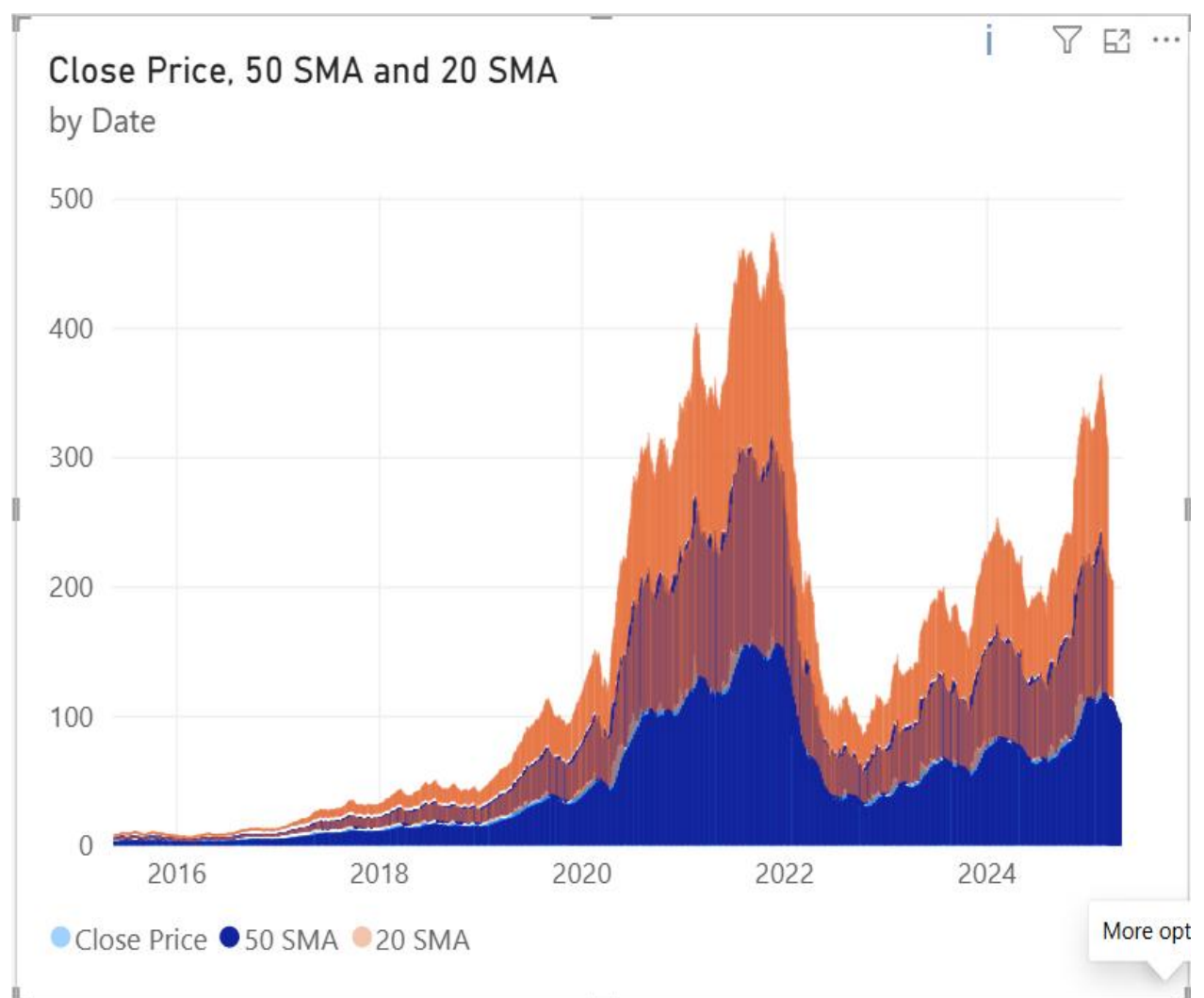
The Date field from the Dim_Date table was used on the X-axis. The X-axis was configured as Continuous to provide a continuous time-series view.

A zoom slider was also enabled to allow detailed analysis of different periods.

6. TRADING VOLUME ANALYSIS

A trading volume chart was created to analyze the number of shares traded over time. The chart uses Date on the X-axis and Total Volume with the Volume 20 SMA as the values.

This visualization helps identify periods of increased or decreased trading activity.



7. VOLUME SPIKE DETECTION

A conditional formatting measure was created to identify significant volume spikes.

The measure compares Total Volume with the 20-Day SMA.

DAX Measure

Volume Color =

IF(

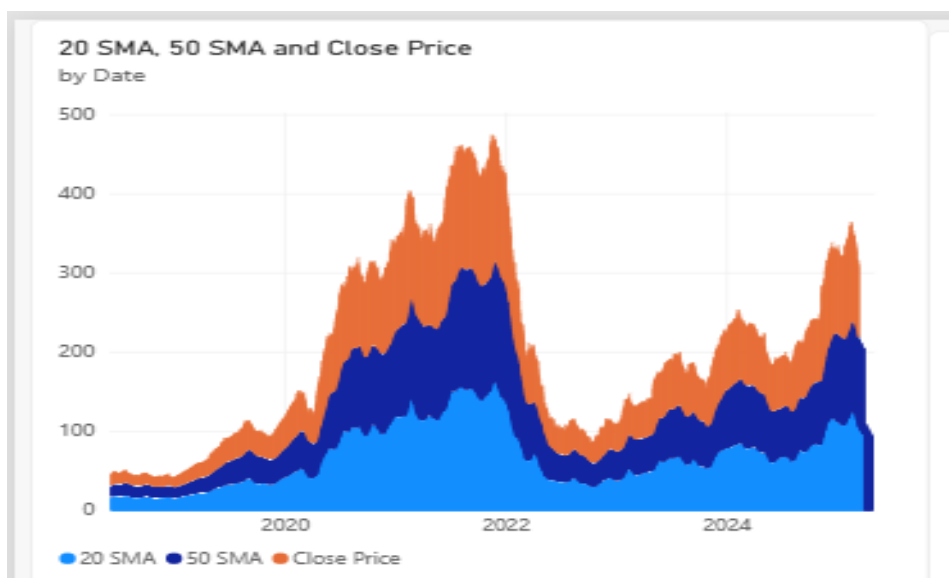
[Total Volume] > [Volume 20 SMA] * 1.5,

"#FF4D4D",

"#26A69A"

)

The conditional formatting was applied to the volume chart using the Field Value option.

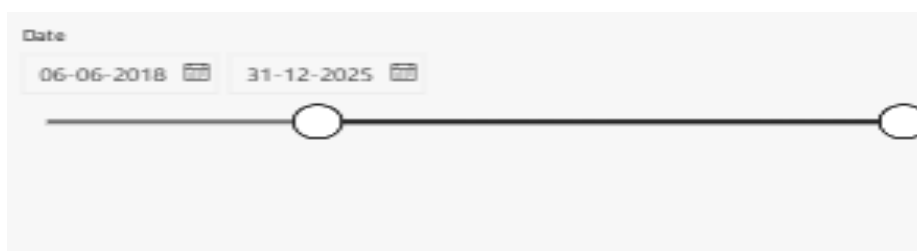


8. DATE RANGE SLICER

A Date Range Slicer was added to make the dashboard interactive.

The Dim_Date[Date] field was added to a Slicer and the slicer mode was set to Between.

Users can select a specific starting date and ending date to analyze a particular period.



9. DYNAMIC STOCK PRICE SELECTOR

A Field Parameter was created using the following price measures:

- Open Price
- High Price
- Low Price
- Close Price

The parameter allows users to dynamically select the stock price metric that they want to display in the chart.



10. CUSTOM TOOLTIP PAGE

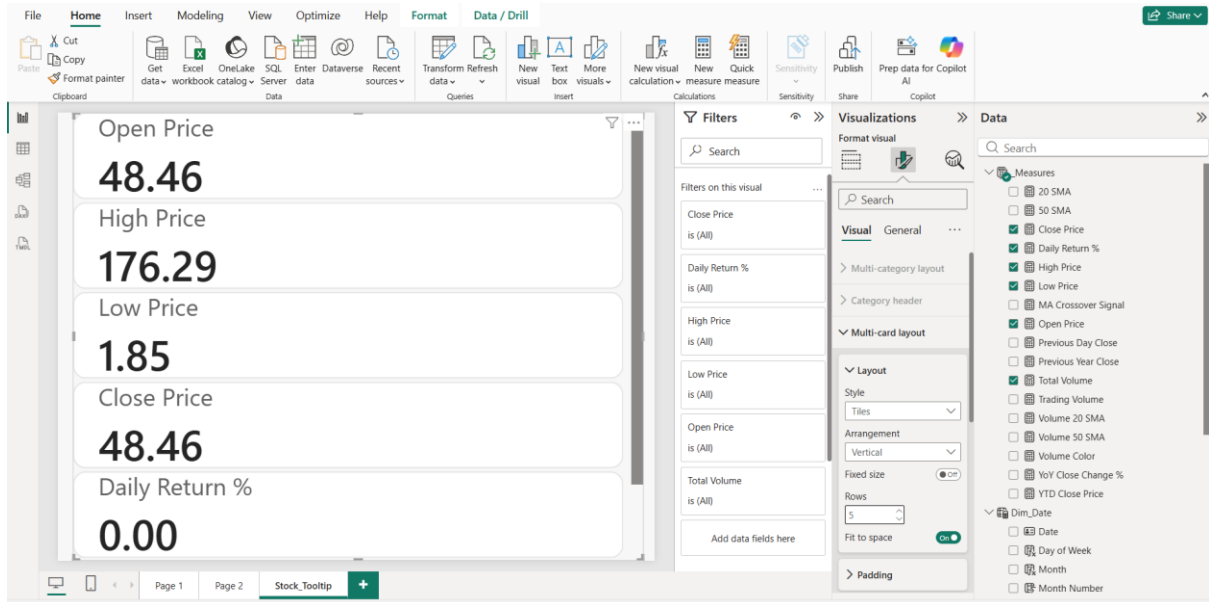
A separate report page named Stock_Tooltip was created to display detailed information when the user hovers over a data point.

The tooltip contains:

- Open Price
- High Price
- Low Price
- Close Price

- Total Volume
- Daily Change %

The page was configured using Allow as tooltip and the canvas type was set to Tooltip.

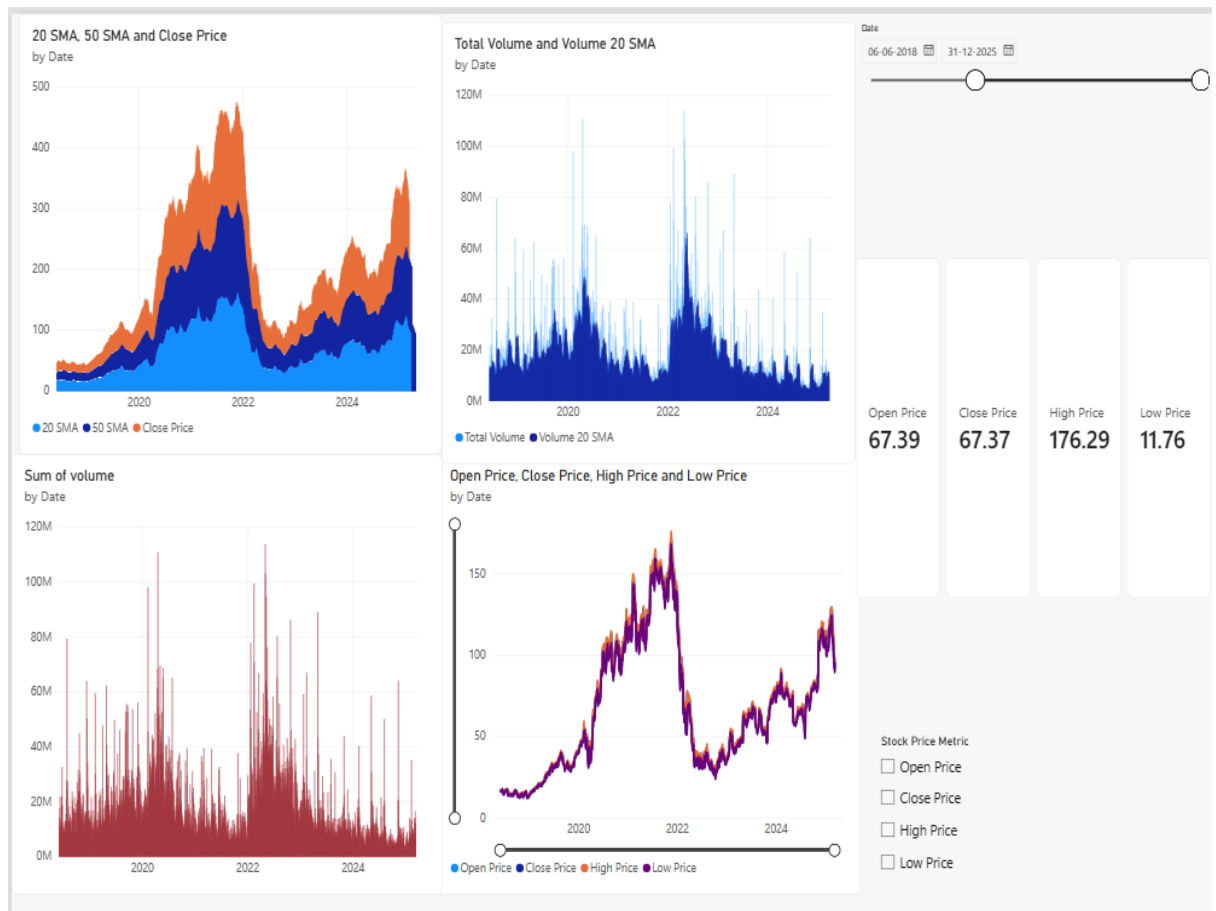


13. FINAL SHOPIFY STOCK EDA DASHBOARD

All the created visualizations and interactive features were combined into a single dashboard.

The dashboard contains:

- Stock price trend
- 20-Day SMA
- 50-Day SMA
- Trading volume
- Volume 20 SMA
- Date range slicer
- Relative date filter
- Dynamic price selector
- Interactive tooltip



Conclusion:

In Week 4, Exploratory Data Analysis was performed on Shopify Stock Data using Microsoft Power BI. The 20-Day and 50-Day Simple Moving Averages were created using DAX to analyze short-term and long-term trading activity. A Moving Average Crossover Signal was developed to compare the two moving averages. A stock price trend chart was created using Close Price, 20 SMA and 50 SMA. Trading volume was analyzed using Total Volume and Volume 20 SMA, and conditional formatting was used to highlight significant volume spikes. Interactive date slicers and a dynamic stock price selector were added to improve dashboard interaction. A custom tooltip page was also created to display detailed stock information when hovering over the chart. Finally, all the visualizations and interactive features were combined into a Shopify Stock EDA Dashboard.